

1. **Sum of First n Integers**

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

2. **Sum of Squares**

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

3. **Sum of Cubes**

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

4. **Sum of Fourth Powers**

$$\sum_{k=1}^n k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

5. **Sum of Fifth Powers**

$$\sum_{k=1}^n k^5 = \frac{n(n+1)(2n+1)(3n^2+3n-1)(4n^3+6n^2-1)}{30}$$

6. **Sum of Odd Numbers**

- Sum of odd numbers $\leq n$:

$$\sum_{\substack{k=1 \\ k \text{ odd}}}^n k = \left[\frac{n+1}{2} \right]^2$$

7. **Sum of Even Numbers**

- Sum of even numbers $\leq n$:

$$\sum_{\substack{k=1 \\ k \text{ even}}}^n k = \left[\frac{n}{2} \right] \left(\left[\frac{n}{2} \right] + 1 \right)$$

8. **Finite Geometric Series**

$$\sum_{k=0}^n a^k = \frac{1-a^{n+1}}{1-a} \quad (a \neq 1)$$

9. **Weighted Geometric Series (Linear)**

$$\sum_{k=0}^n k a^k = \frac{a[1-(n+1)a^n + na^{n+1}]}{(1-a)^2}$$

10. **Weighted Geometric Series (Quadratic)**

$$\begin{aligned} & \sum_{k=0}^n k^2 a^k \\ &= \frac{a[(1+a) - (n+1)^2 a^n + (2n^2 + 2n - 1)a^{n+1} - n^2 a^{n+2}]}{(1-a)^3} \end{aligned}$$

11. **Binomial Theorem**

$$\sum_{k=0}^n \binom{n}{k} = 2^n$$

12. **Alternating Binomial Sum**

$$\sum_{k=0}^n (-1)^k \binom{n}{k} = 0$$

13. **Weighted Binomial Sum**

$$\sum_{k=0}^n k \binom{n}{k} = n \cdot 2^{n-1}$$

14. **Harmonic Series**

$$\sum_{k=1}^n \frac{1}{k} = H_n \approx \ln n + \gamma$$

15. **Sum of Reciprocal Squares**

$$\sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}$$

16. **Fibonacci Series Sum**

$$\sum_{i=1}^n F_i = F_{n+2} - 1$$

17. **Telescoping Series**

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1}$$

18. **Sine Taylor Series**

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} = \sin x$$

19. **Cosine Taylor Series**

$$\sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} = \cos x$$

20. **Exponential Taylor Series**

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

21. **Geometric Series (Base 2)**

$$\sum_{k=1}^n 2^k = 2^{n+1} - 2$$

22. Finite Geometric Series (General)

$$\sum_{k=1}^n x^k = \frac{x(x^n - 1)}{x - 1}, \quad x \neq 1$$

$$\sum_{k=1}^n x^k = \frac{1 - x^{n+1}}{1 - x}, \quad x \neq 1$$

23. Weighted Geometric Series (Base 2)

$$\sum_{k=1}^n 2^k \cdot k = 2^{n+1} \cdot n - 2$$

24. Exponential Series with Constant Exponent

$$\sum_{k=1}^n 2^{km} = \frac{2^m(2^{mn} - 1)}{2^m - 1}, \quad m \text{ constant}$$